Assignment 3
White-box testing

Assignment Objectives:

- Generating test cases based on white box testing.
- Use JUnit for implementing the test cases.
- Use Jenkins for executing the test cases

Prerequisite:

- Create a team –5 or 6 members.

Assignment = In-Class assignment + Take-Home assignment

Before Lab3: Docker-Jenkins-Gitea–AllureReport - install on your computer
See ***Info_CI_CD.pdf* tutorial.**

- **In-Class assignment**
  - First hour of the laboratory.
  - Maxim 50 XP
  - TO DO:
    1. **Design test cases (WBT)** (20 XP)
       - Statement coverage or Condition coverage or Path coverage or other coverage criterion
         - Minimum 5 test cases will be created for In-Class Problem below.
       - Use the template document: Lab03_WBT_Form.xls.
    2. **WBT Tests** (20 XP)
       - Create a Maven project. Use the Tutorial from Lecture 2.
       - Implement the test cases that you have designed and test your implementation.
    3. **[Jenkins]** (10 XP)
       - Add the project to git using gitea. Use the Tutorial from Lecture 2.
       - Creating a pipeline job in Jenkins for executing the test cases.
    4. **In-class Problem**: Problem ***Set*** (test the methods ***AddAValue*** and ***IsInTheSet***) is found in the archive Lab03Files.
- **Take-Home assignment**
  - At home.
  - Maxim 100 XP
  - TO DO:
    1. **Design test cases (WBT)** (30 XP)
       - All coverage criteria (all statements, decision/condition, paths, loops) for the Take-Home Problem: excel file with the design of the test cases + Java implementation
    2. **BBT Tests** (30 XP)
       - Create a Maven project. Use the Tutorial from Lecture 2.
       - Implement the test cases that you have designed and test your implementation.
    3. **[CI_CD]** (40 XP)
       - Add the project to git using gitea. Use the Tutorial from Lecture 2.
       - Creating a pipeline job in Jenkins for executing the test cases.
  - **Take-Home Problem**: Problem ***GiveNextDate*** is found in the archive Lab03Files.
- **Bonus assignment PbSet (+50XP)**
  - At home but before 3 April 2026, hour 20:00.
  - See information on the study in assignment BonusAssignment_Lab3_PbSet.

Assignment and Delivery date:

1. Assignment date: laboratory 3
2. Delivery date for **In-Class assignment:** laboratory 3 (max 50 XP)
3. Delivery date (first) **for Take-Home assignment**: laboratory 4 (max 100 XP)
4. Delivery date (last) **for Take-Home assignment**: laboratory 5 (max 50XP)

Turn in:
Delivered and presented in class AND upload on Teams (only one student from the team) after Delivery in class the following archive:

a. Lab3_IC_TH_93X_Name01Name02.zip
   i. In-class problem
      1. Lab03_WBT_Form.xls (Statement coverage or Condition coverage or Path coverage or other coverage criterion).
      2. Source code: Implementation of minimum 5 test cases.
      3. Screen capture with the Gitea and Jenkins and Allure report of your tests.
   ii. Take-Home problem
      1. Lab03_WBT_Form.xls (CFG + CC + Paths using, test cases design for: all statements, decisions/conditions, paths, loops coverage criteria). **Complete analysis of WBT!**
      2. Source code: Implementation of the test cases.
      3. Screen capture with the Gitea and Jenkins and Allure report of your tests.
   iii. document with the list of members. No "guest name"! Every person mentioned as a collaborator must contribute.

b. Lab3_Bonus
   i. See information on what to submit in assignment BonusAssignment_Lab3_PbSet.
   ii. This bonus assignment must be submitted by 3 April 2026, hour 20.00.

References

- See Lecture 3, Seminar 3
- See documents for Lab03Files.zip

Leap-year bug (Bernd Bruegge Allen H. Dutoit, Object-Oriented Software Engineering Using UML, Patterns, and Java, Pearson Education Limited, 2014)
A supermarket was fined $1000 for having meat around 1 day too long, on February 29, 1988.
The computer program printing the expiration date on the meat labels did not take into account that 1988 was a leap year.

Other Leap-Year Problems
https://www.csl.sri.com/users/neumann/cal.html